

Olist Brazilian E-Commerce

Driving Revenue Growth & Customer Retention

An End-to-End Business Intelligence Case Study



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Project Overview

Developed an end-to-end Business Intelligence solution using the Olist Brazilian E-Commerce dataset to analyze revenue, customer behavior, product performance, operational efficiency, and customer satisfaction through interactive Power BI dashboards.

Project Objective

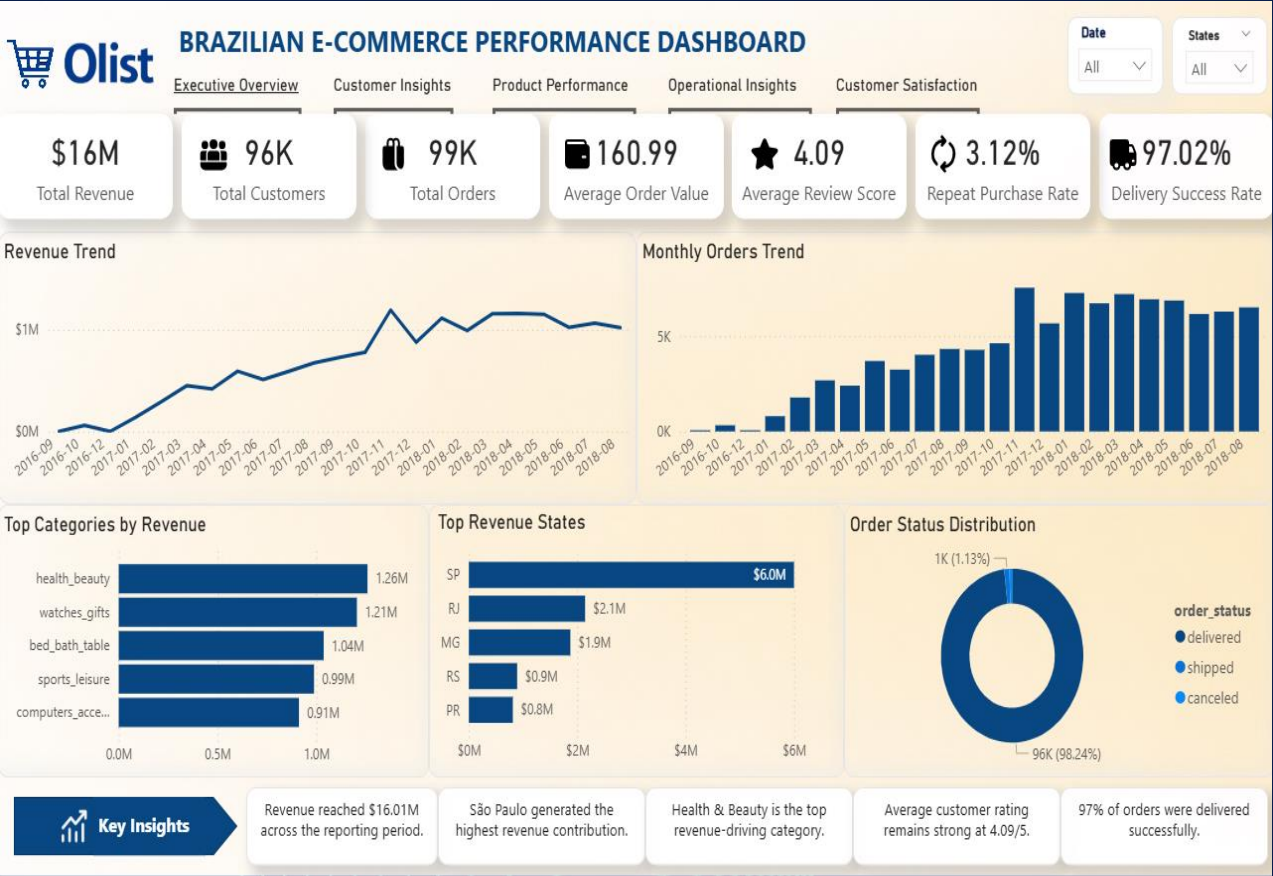
Build a centralized analytics solution that helps business stakeholders monitor key performance metrics, identify business opportunities, and support data-driven decision-making.

Business Problem

Business stakeholders lacked a centralized view of revenue, customers, products, operations, and customer satisfaction, making it difficult to monitor business performance and identify improvement opportunities.

Business Outcome

Built an interactive Power BI dashboard that provides actionable insights across revenue, customers, products, operations, and customer satisfaction to support better business decisions.



Business Understanding

Business Context

Olist is a Brazilian e-commerce marketplace that generates large volumes of transactional data across customers, products, sellers, payments, and deliveries. Analyzing this data helps businesses understand performance, identify improvement opportunities, and support better decision-making.

Business Objective

Provide stakeholders with a centralized analytics solution to monitor business performance, identify improvement opportunities, and support data-driven decision-making.

Project Vision: Transform transactional data into actionable business insights that improve strategic decision-making.



Revenue



Customers



Products



Operations



Customer Satisfaction

Stakeholders & KPIs

Key metrics monitored by each business stakeholder.



Executive Management

- **Total Revenue**
- Total Orders
- Total Customers
- Average Order Value



Product Team

- **Top Revenue Category**
- Product Revenue
- Category Orders



Sales & Marketing

- **Repeat Purchase Rate**
- Revenue per Customer
- Monthly Customer Growth



Operations Team

- **Average Delivery Days**
- Delay Rate
- Cancellation Rate



Customer Experience Team

- **Average Review Score**
- Five-Star Review Rate
- Total Reviews

The dashboard was designed to provide role-specific insights, enabling each stakeholder to make faster and more informed business decisions.

Dataset Overview



96K+
Customers

99K+
Orders

33K+
Products

9
Tables

Dataset Summary

This project uses the Olist Brazilian E-Commerce Public Dataset, which contains real marketplace transactions from 2016–2018. The dataset includes customers, orders, products, payments, sellers, deliveries, and customer reviews for end-to-end business analysis.

Key Facts

- Marketplace: **Olist (Brazil)**
- Time Period: **2016–2018**
- Dataset Type: **Transactional E-Commerce**
- Source: Kaggle Public Dataset

Core Tables Used

- ✓ Customers
- ✓ Orders
- ✓ Order Items
- ✓ Payments
- ✓ Products
- ✓ Sellers
- ✓ Reviews
- ✓ Category Translation
- ✓ Geolocation table

Data Model & Star Schema

Data Model Overview

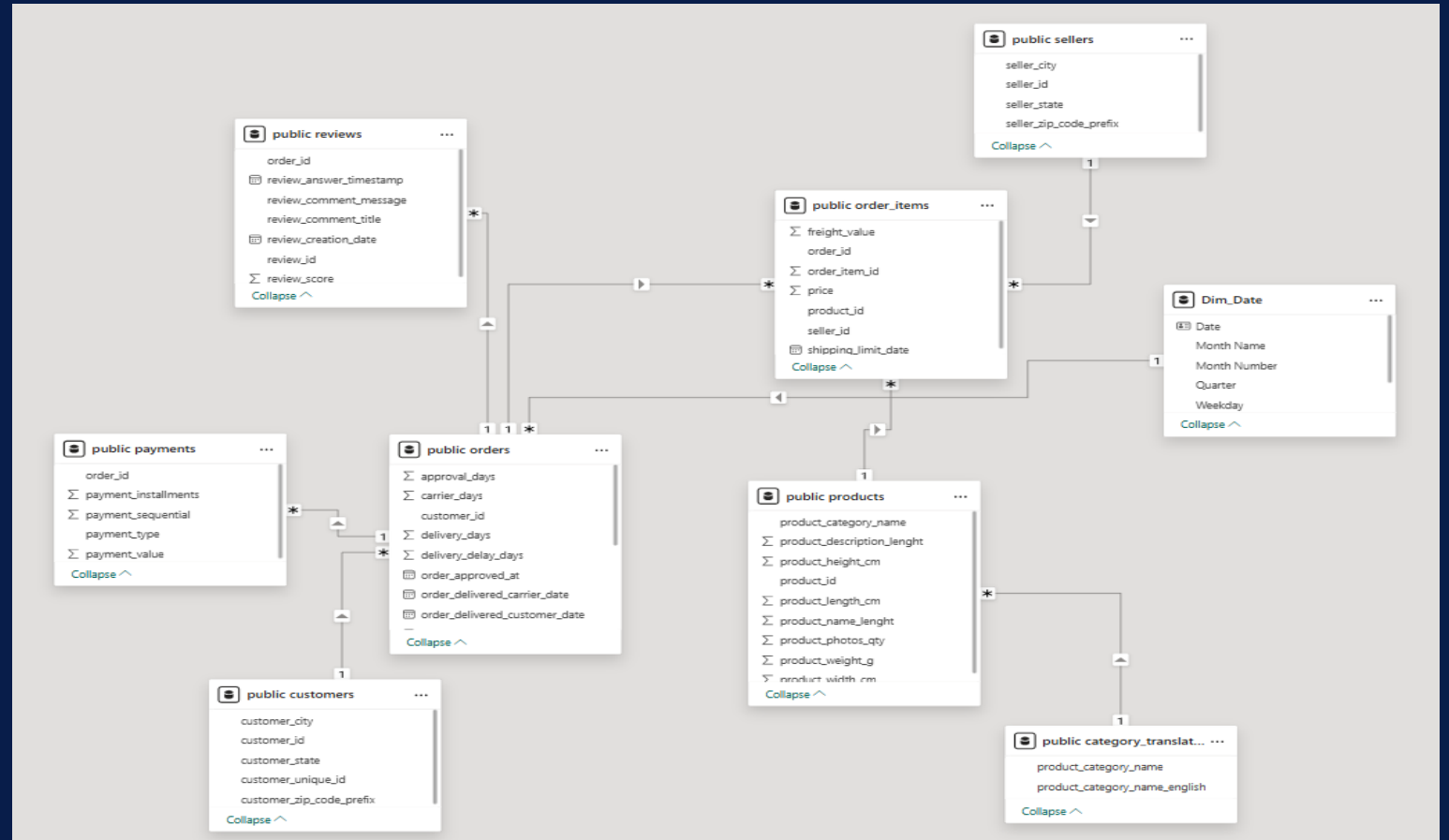
A star-schema-inspired data model was created to organize transactional and dimension data for efficient reporting. One-to-many relationships with single-direction filtering were used to improve performance and simplify analysis.

Fact Tables

- Orders
- Order Items
- Payments
- Reviews

Dimension Tables

- Customers
- Products
- Sellers
- Category Translation
- Dim Date



The data model provides a scalable foundation for accurate calculations, faster reporting, and consistent business insights.

End-to-End Analytics Workflow

Complete data pipeline from raw data to business insights



Project Outcomes

- ✓ Cleaned Dataset
 - ✓ Feature Engineered Dataset
 - ✓ SQL Business Analysis
 - ✓ Star Schema Data Model
 - ✓ DAX Measures
 - ✓ Interactive Dashboard
 - ✓ Business Insights
 - ✓ Strategic Recommendations
-

Executive Overview Dashboard

A five-page interactive Power BI dashboard designed to monitor overall business performance across revenue, customers, products, operations, and customer satisfaction.

Revenue

Track sales performance and revenue trends

Customers

Analyze customer growth and repeat purchases

Products

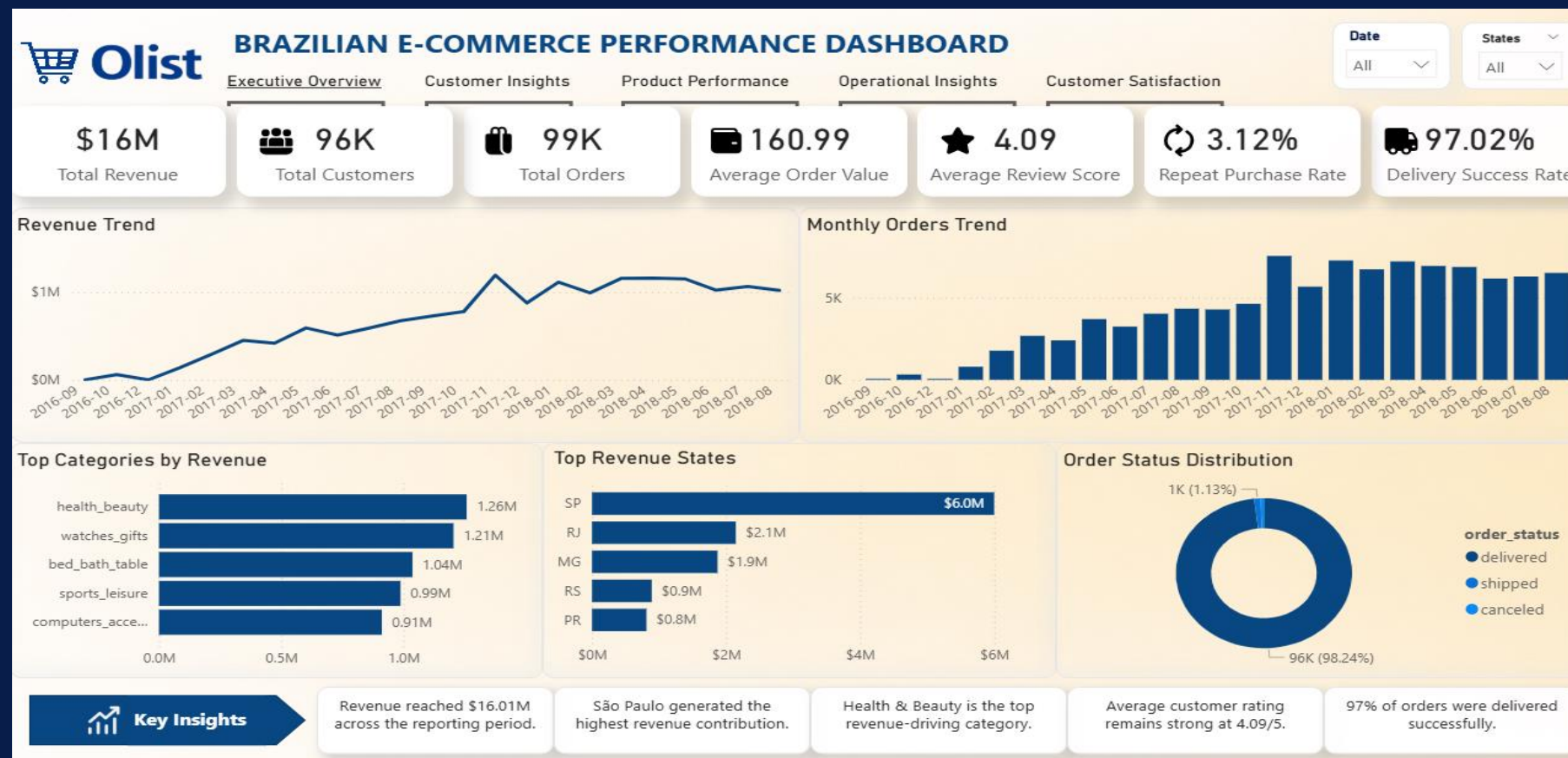
Identify top-performing products and categories

Operations

Monitor delivery performance and operational efficiency

Customer Satisfaction

Measure review scores and customer sentiment



Interactive Power BI solution designed for business decision-making.

Customer Insights

Analyze customer acquisition, regional distribution, purchase behavior, and retention performance.

Purpose

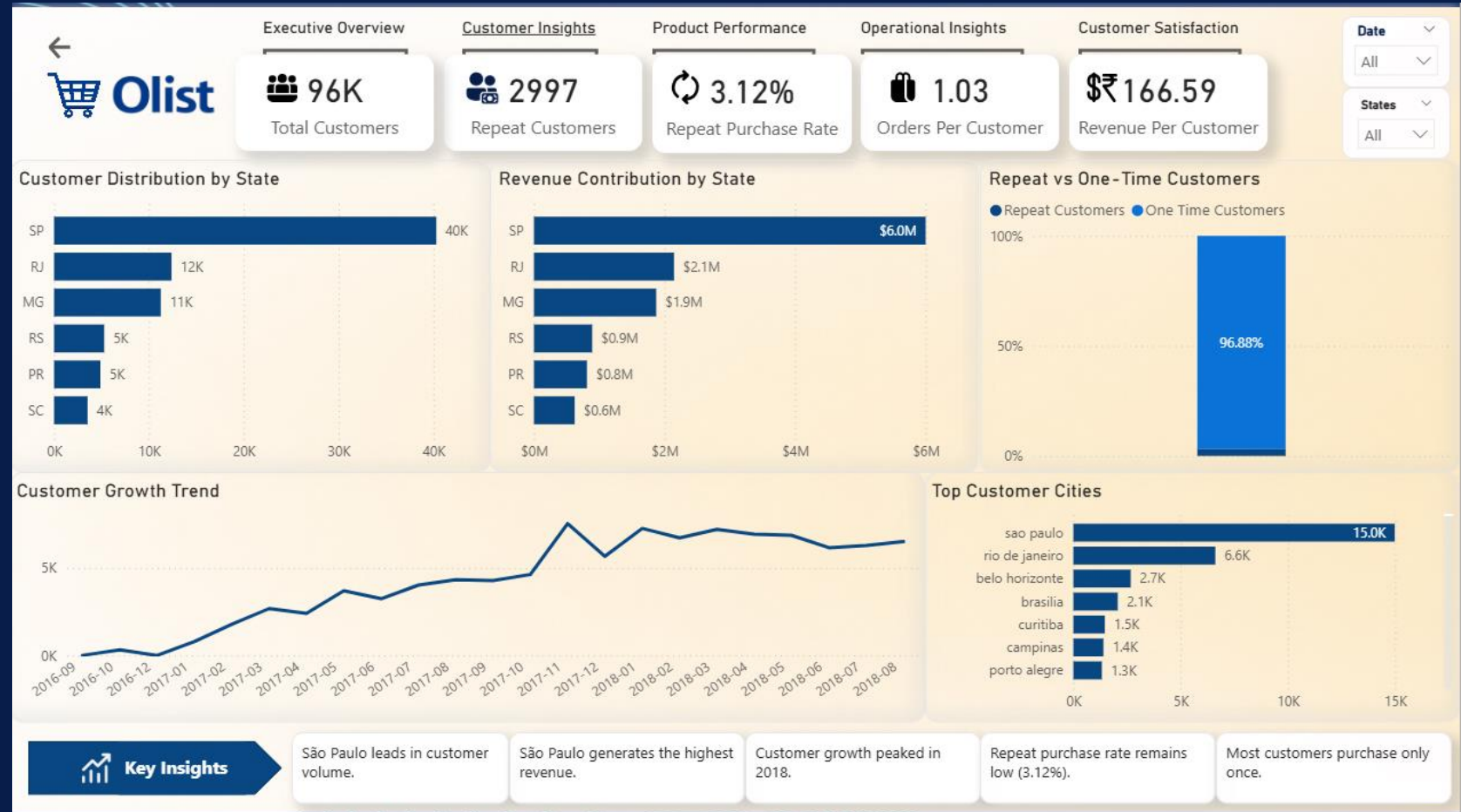
- Understand customer distribution across regions.
- Analyze customer acquisition and retention.
- Identify repeat purchase behavior.

Key KPIs

- Total Customers
- Repeat Customers
- Repeat Purchase Rate
- Revenue per Customer
- Orders per Customer

Key Insights

- São Paulo has the largest customer base.
- Customer growth increased during 2018.
- Repeat purchase rate remains low at **3.12%**.
- Most customers purchase only once.



Supports customer retention strategies, regional expansion, and targeted marketing decisions.

Product Performance

Analyze customer growth, purchase behavior, regional distribution, and retention performance.

Purpose

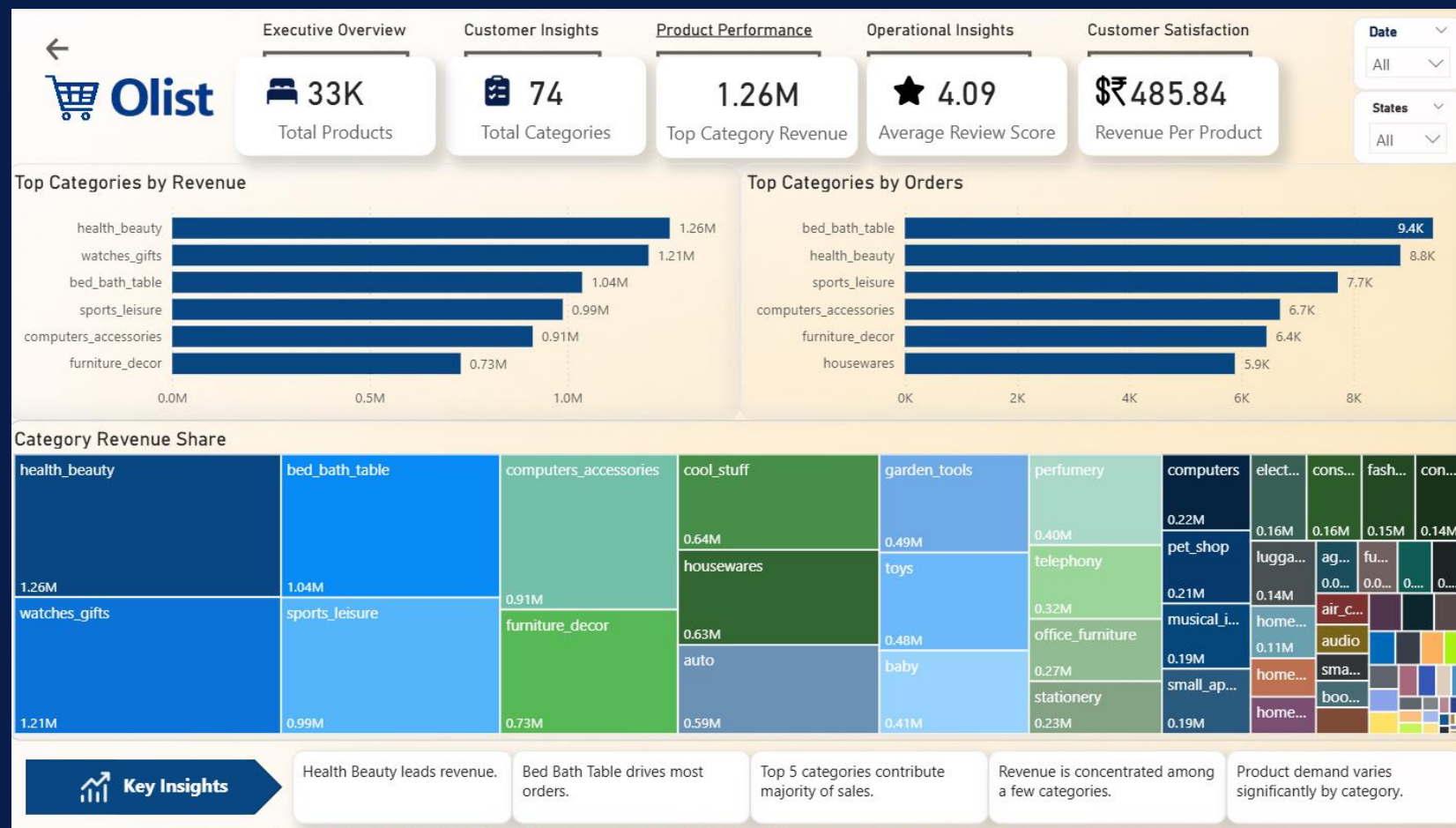
- Evaluate category-level sales performance.
- Identify high-performing product categories.
- Analyze revenue distribution across products.

Key KPIs

- Total Products
- Total Categories
- Top Category Revenue
- Revenue per Product
- Average Review Score

Key Insights

- Health & Beauty generated the highest revenue.
- Bed Bath Table recorded the highest order volume.
- Top five categories contribute a major share of total revenue.



Supports product portfolio optimization, inventory planning, and category-level business decisions.

Operational Insights

Monitor order fulfillment, delivery performance, delays, and operational efficiency.

Purpose

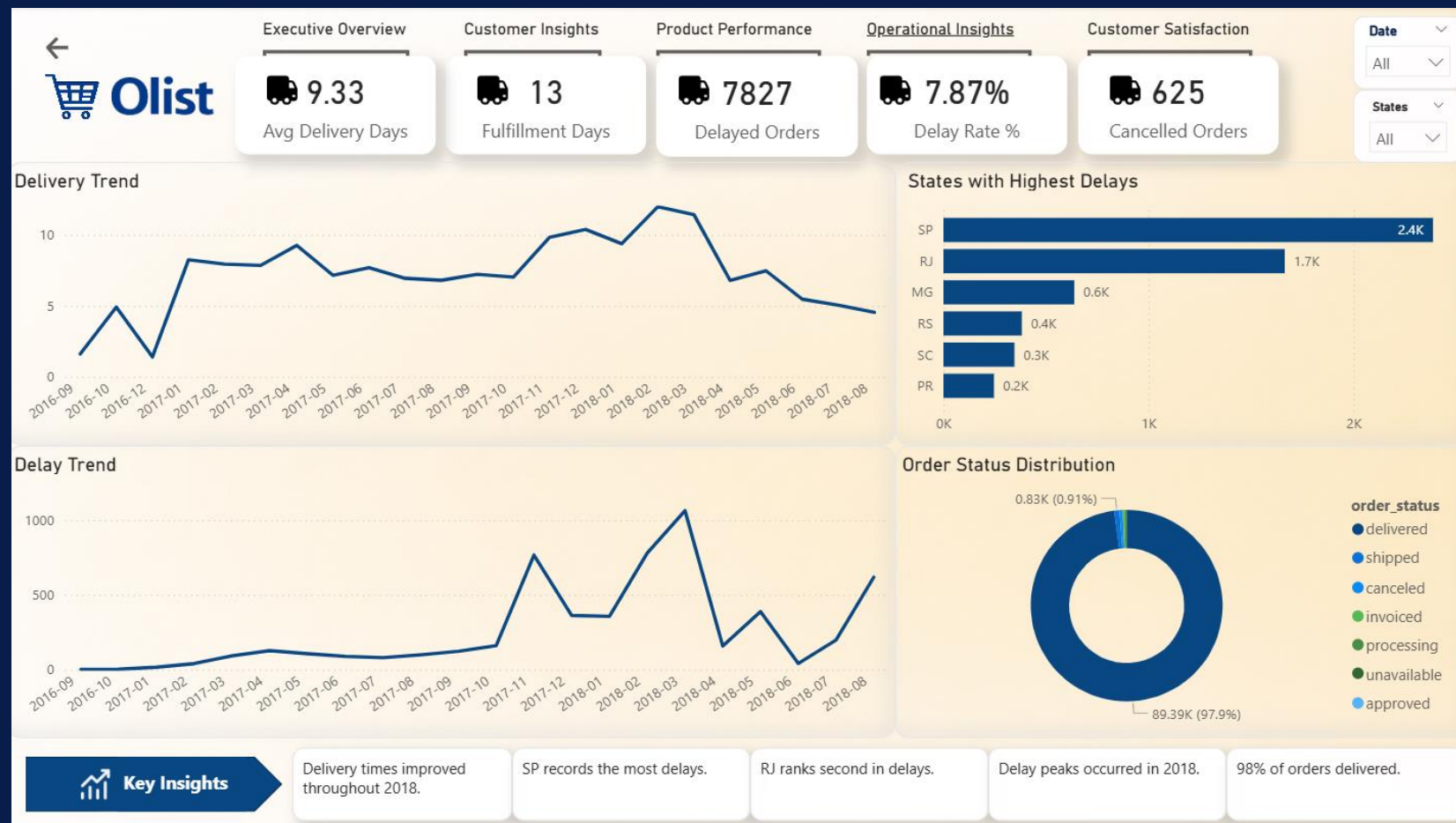
- Evaluate delivery and fulfillment performance.
- Monitor delayed and cancelled orders.
- Identify operational bottlenecks across regions.

Key KPIs

- Average Delivery Days
- Fulfillment Days
- Delayed Orders
- Delay Rate
- Cancelled Orders

Key Insights

- Average delivery time was **9.33 days**.
- Delay rate reached **7.87%** of total orders.
- São Paulo recorded the highest delayed order volume.
- Delivery success rate remained above **97%**.



Supports logistics optimization, operational efficiency, and delivery performance improvement.

Customer Satisfaction

Measure customer sentiment through review scores, ratings, and satisfaction trends.

Purpose

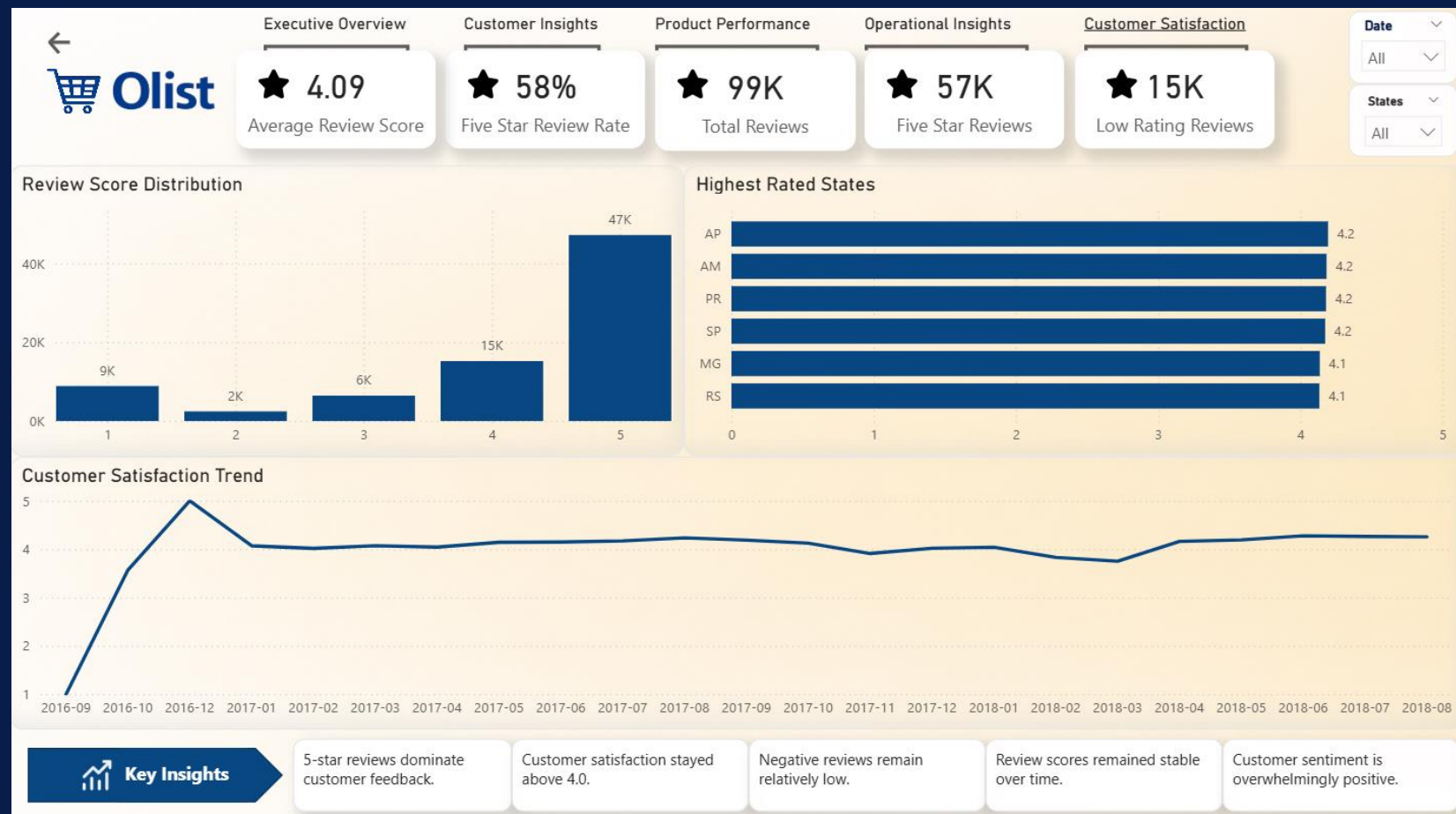
- Evaluate overall customer satisfaction.
- Analyze review ratings and customer feedback.
- Monitor satisfaction trends over time.

Key KPIs

- Average Review Score
- Five-Star Review Rate
- Total Reviews
- Five-Star Reviews
- Low Rating Reviews

Key Insights

- Average review score remained at **4.09 out of 5**.
- **58%** of reviews were five-star ratings.
- Customer satisfaction remained stable over time.
- Overall customer sentiment remained highly positive.



Supports service quality improvement, customer experience monitoring, and long-term customer loyalty.

Business Insights & Strategic Recommendations

The analysis highlights strong revenue growth, high customer satisfaction, and efficient delivery performance, while identifying customer retention and revenue concentration as the primary opportunities for improvement.

Key Business Insights

- Revenue reached **R\$16.01M** with steady business growth.
- Customer retention remains low with a **3.12%** repeat purchase rate.
- Revenue is concentrated in a small number of product categories.
- **7.87%** of orders experienced delivery delays despite a **97%** delivery success rate.
- Customer satisfaction remained strong with an average review score of **4.09/5**.

Strategic Recommendations

- Launch loyalty and personalized marketing campaigns to improve repeat purchases.
- Expand investment in high-performing product categories while growing underperforming categories.
- Optimize logistics in high-delay regions to reduce delivery delays.
- Use customer feedback to continuously improve service quality.
- Monitor KPIs regularly through interactive Power BI dashboards.

The solution enables stakeholders to monitor business performance, identify growth opportunities, and make faster, data-driven business decisions.

Thank You

Presenter

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<https://github.com/Mslp18>